

Linked Open Data for Comic Books

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Abstract

Linked Open Data (LOD) is a collective effort by researchers, data scientists, and the public with a vision of a fully connected world. Metadata is the key to Linked Open Data (LOD), but various groups, institutions, and projects use various formatting standards for their data that can make it inaccessible across the web for other users. Not only does this cause information silos, but creates confusion and roadblocks when encoding and indexing new data. Comic books (including graphic novels and manga) are an exemplar of these issues. Comic books differ from a standard book that may be indexed as part of a library or archive's collection for several reasons, mainly 1) the visual structure and 2) publication and serialization conventions. This paper will explore various ontologies, tools, and frameworks available to index comics and the current research and possibilities for making comic book metadata both linked and open.

Keywords: linked open data, comic books, ontologies, digital libraries, Wikidata

Introduction

Comics or any form of sequential, visual storytelling are not new mediums for artists and writers, but their acceptance as formative, engaging, and mature materials has only become recognized at an academic level in the past few decades (Duncan & Smith, 2009). Comics are more focused on storytelling rather than evoking emotion or capturing a moment in a stand-alone work. Using sequential art as a defining aspect, works dating far before the printing press can be considered and analyzed as comics. The Bayeux Tapestry from the 11th century recounts events surrounding the conquest of England by the Duke of Normandy through a linear series of sequential images. The 70-meter tapestry contains 58 scenes with hundreds of characters (and horses). Rather than a singular image to remark on this period of history, the sequence of scenes creates a historical account that has a lasting impact to this day, even inspiring embroiderers from the 19th century to undertake the reconstruction process. Pre-Columbian pictographs, Greek and Roman friezes, and ancient Egyptian wall paintings are further examples of sequential art that archeologists, historians, and creatives read and analyze

to learn more about these cultures and societies. While defining these works as “comics”, especially compared to modern examples, is up for debate, the importance and longevity of sequential art cannot be ignored.

However, broadening the definition of comics to include these historic works increases the visibility of works labeled as comics and the validity of the medium as a place for academic research. Comic books fall under the comics umbrella, but differ from the historical examples presented above. Comic books, in the more modern sense, can be traced back to the late 19th century when Rodolphe Töpffer published his innovative picture-stories: oblong albums of sequential art with “slapstick tales of absurdist antiheroes who struggled against the whims of fate”. The term “comic” is derived from early comic strips, which were inspired by Töpffer, in newspapers that would later be copied into volumes to create comic *books* (Duncan & Smith, 2009). While many of these strips were comical in subject, comic book subjects have matured and expanded to encompass many genres, from autobiographical to horror. Art Spiegelman's *Maus*, which is based on his father's experience as a Jewish man under the Nazi regime, was the first graphic novel (another term for comic books that was more pleasing to publishers and academics) to be awarded the Pulitzer Prize in 1992, to work was so impressive that academics hesitated to call it a graphic novel. Comic books are a valuable scholarly and cultural resource, and their unique structure can be supported and shared by descriptive metadata frameworks.

Comic Structure and LOD Potential

As defined by Scott McCloud, comics are “juxtaposed pictorial and other images in deliberate sequence, intended to convey information and/or to produce an aesthetic response in the viewer” (McCloud, 1993, p.9). Comics tell stories through sequential art, text, and visual juxtaposition. The physical space created between panels and images is called the “gutter”, a place where the reader makes connections between sequences. The act of passing one's eyes over the gutter between sequences is similar to how viewers fill in the gaps between film frames to experience what feels like a moving image. What makes the gutter stand out is that it is where

the reader interjects their own experience, emotions, and perceptions to form the “missing” parts of the story in their minds. The gutter is an example of the unique, visual aspects of comics that can be encoded with descriptive metadata. Comics do not need text to be comics and many will have a series of text-less sequences, so the visual information conveyed by the artist through their art and sequencing choices is paramount to interpreting the story and potentially linking to other data.

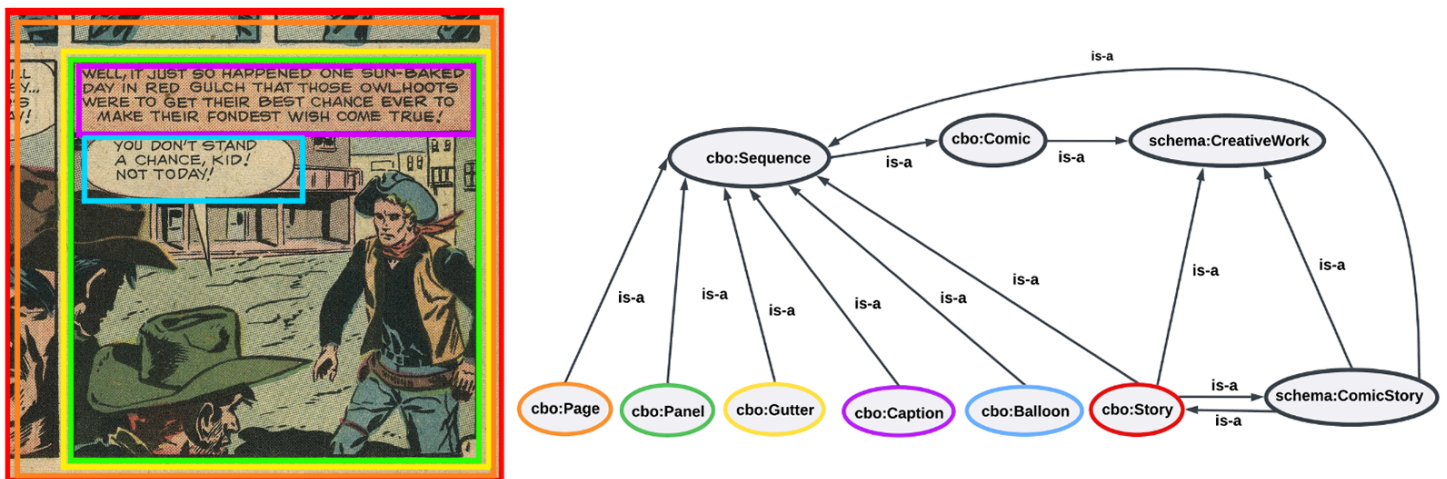


Fig. 1. A recreation of the Comic Book Ontology (CBO) Sequence model designed by Sean Petiya mapped to an excerpt from *Billy the Kid*, issue #13 (1958, September). “I’ll Outdraw Him!”. Charlton Comics. Public Domain. <https://digitalcomicmuseum.com/index.php?dclid=26256>

For LOD to work, following established standards and vocabularies is necessary to encode human and machine-readable metadata. The Resource Description Framework (RDF) provides a data model that allows schemas to work harmoniously together to describe data and aid in its discoverability. The RDF triple and identifiers like Uniform Resource Identifiers (URIs) and web addresses allow users and their machines to read and connect with resources across the web and create expansive knowledge graphs. The eXtensible Markup Language (XML) is a metalanguage format recommended by the World Wide Web Consortium (W3C) to structure the metadata in a way that’s readable for both humans and machines. RDF and XML are not required for data to be LOD, but are widely adopted.

A singular page of a comic can contain a mix of panels, captions, gutters, and text balloons (see Fig. 1). Sean Petiya addresses these narrative elements with the Comic Book Ontology (CBO), a W3C Web Ontology Language (OWL) ontology and RDF metadata vocabulary (Petiya, 2023). There are other schemas specific to comic books, such as ComicsML, an XML-based markup language designed specifically for digital comics, and Comic Book Markup Language (CBML), a Text Encoding Initiative (TEI) based XML vocabulary for encoding and analyzing comic books. Petiya developed CBO not as a replacement, but as an addition to other schemas to semantically enrich new and existing comic book data. ComicsML's atomic level element is the panel, so as to focus on the sequence of panels within a comic (McIntosh, 2001). CBML goes further to incorporate dialogue balloons and captions with an established vocabulary that covers various elements that could appear in a comic book, such as advertisements and news features (Walsh, 2012).

Petiya describes how a descriptive ontology specific to the layout standards of comic books can semantically enrich research and knowledge in other fields, specifically, mental health and wellness (Petiya, 2023). Petiya uses graphic medicine, a sub-genre of comics that covers healthcare information, as a focus for applying an LOD approach to comics and graphic works. In conjunction with Library of Congress Subject Headings (LCSH), Medical Dictionary for Regulatory Activities Terminology (MedDRA), and the National Cancer Institute Thesaurus (NCIT), Petiya describes sequences within a comic book, *States of Mind*, written by a patient with bipolar disorder (lcsch:sh85080541) and their father/caregiver. Petiya's CBO sequence model adds nodes for pages (cbo:Page), panels (cbo:Panel), and gutters (cbo:Gutter) connected by the schema:about property to mental health topics, experiences, and related symptoms such as stress (meddra:19942209), panic attacks (meddra:10033664), loss of appetite (ncit:C113630), and involuntary commitment (meddra:10080523). Enhancing and extending the description of comics, such as those that include sequences where mental health and wellness are featured, can destigmatize comics and categorize them as a legitimate

medium that can encompass mature themes.

Comic Publication and Incomplete Data

Manga, which are comics and graphic novels originating from Japan, are now globally recognized and have been increasing in popularity since the 1990s. However, due to their subculture associations, they do not have robust authority data within libraries to address their publication methods. Manga are usually first published in magazines, either page-by-page or chapter-by-chapter, and later republished as a monograph or series of monographs if received well. Western comics are similarly complex, with one or many issues being published and many longstanding series employing multiple artists, writers, and contributors. For a reader, manga and comic books are identified as distinct intellectual works that can contain a multitude of chapters, volumes, and stories. The Functional Requirements of Bibliographic Records (FRBR) model, a conceptual entity-relationship model developed by the International Federation of Library Associations and Institutions (IFLA), defines four levels of representation to support user retrieval and access in online library catalogs and bibliographic databases. These levels are work, expression, manifestation, and item, separate from standards such as Anglo-American Cataloguing Rules (AACR) or International Standard Bibliographic Description (ISBD). FRBR can be utilized for cataloguing manga and comic books as a *work*, the distinct intellectual or artistic creation, instead of a *manifestation*, the physical embodiment of a work, or an *item*, a single example of a manifestation, like a library may do for a standard book (Librarianship Studies & Information Technology, 2019).

In their 2013 paper, He, Mihara, Nagamori, and Sugimoto come to the conclusion that appropriate work instances and relationships from bibliographic records for manga can be created using DBpedia, a community LOD project that helps users query data from Wikipedia, given that there are no issues in the source records and the source already exists. These researchers used DBpedia to group various instances of manga works that have been translated into multiple languages by creating a record-set of title, author, and language. For example, the

record-set of manga title “Siki”, author “Yokoyama”, and language code “ja” for Japanese matches with title “Shiji”, author “Yokoyama”, and language code “zh” for Chinese. These were identified as separate works in the bibliographic data found using DBpedia. Once identified, these record-sets could be compiled into a singular identifiable work. The authors focused on linking manga works with varied language expressions, instead of graphic or textual expressions. They state that information on graphic expression is not yet straightforward enough in LOD resources to be used as authority information (He, Mihara, Nagamori, & Sugimoto, 2013).

Compiling, writing, and finding LOD for comic books is an ongoing endeavor taken on by many. Michigan State University (MSU), which houses one of the largest comic book collections in North America, developed the Comics as Data North America (CaDNA) project to extract and clean MARC records describing comics published in North America. Part of their research involved analyzing the Library of Congress Classification publisher and author fields, where they discovered large discrepancies due to a lack of authority control for publishers in the catalogue. For example, works published by Marvel may be described as being published by “Marvel Comics”, “Marvel Publishing”, “Marvel Comics Group”, and other such names. Similar to how He, Mihara, Nagamori, and Sugimoto linked manga titles and authors with different language versions, the CaDNA group used OpenRefine’s Wikidata Reconciliation Service to connect various records for Marvel published comics into one item: Marvel Comics (Q173496). The process revealed that around 10,000 publishers were missing in Wikidata. They performed the same process for authors and found that many items were incorrectly matched to authors with similar names. In October 2020, the MSU Department of English Graphic Possibilities research Workshop hosted a virtual Wikidata edit-a-thon to address and add missing comic book publishers and authors from 1929 to 1956 using information derived from MSU’s Comic Art Collection metadata. The group found that many of the inconsistencies in the MSUL catalogue stemmed from greater inconsistencies in authority files in OCLC that were beyond the scope of

their project (Topham, K., Chambliss, J., Wigard, J., & Huff, N., 2022). These projects showcase the need for greater communication between comic book cataloguers and LOD resource creators.

Conclusion

This paper only scratches the surface of the complexity involved in developing LOD for comics. While there are markup languages that exist to support encoding metadata for the unique visual structure and publishing standards of comics, they are not widely implemented, and even if they were, much of the basic data for comic books is missing from open databases. As with many LOD projects and digital databases, ample time and community input are necessary to upload, create, and encode data. The examples of comics presented in this paper do not encompass born-digital comics, but since the early 2000s, the growth of digitally born comics has exploded, and their forms and delivery methods have evolved rapidly. A combination of schemas like CBO, ComicML, and CBML, utilized by university libraries and community databases, possibly through an event like the MSU hackathon, could bolster new ideas for comic book LOD and garner attention from the semantic web, academic, and comic-book-reader communities.

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